

Boss Challenge — Paper 16 (Class 7)

Light | The Triangle & its Properties | Wastewater Story | Comparing Quantities

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. In a clear, uniform medium, light from a source travels along:
(A) spreading circles
(B) straight lines
(C) zig-zag paths
(D) curved paths
2. The sum of the three interior angles of any triangle is always:
(A) 180°
(B) 90°
(C) 270°
(D) 360°
3. The used, dirty water released from homes, offices and factories is called:
(A) groundwater
(B) rainwater
(C) distilled water
(D) wastewater
4. A ratio compares two quantities of the same kind by:
(A) subtraction
(B) division
(C) multiplication
(D) addition
5. When a ray of light strikes a plane mirror, the angle of incidence is always:
(A) always ninety degrees
(B) half the angle of reflection
(C) twice the angle of reflection
(D) equal to the angle of reflection
6. In a triangle, two angles measure 50° and 60° . The third angle is:
(A) 70°
(B) 80°
(C) 110°
(D) 60°
7. The wastewater carried away in drains from kitchens, toilets and bathrooms is specially called:
(A) sludge
(B) sewage
(C) clarified water
(D) manure

8. The ratio 15 : 25 written in its simplest form is:
- (A) 5 : 3
 - (B) 15 : 25
 - (C) 1 : 2
 - (D) 3 : 5
9. The image you see of yourself in a flat bathroom mirror is:
- (A) real and the same size
 - (B) virtual, inverted and larger
 - (C) real, inverted and smaller
 - (D) virtual, erect and the same size as you
10. An exterior angle of a triangle is equal to the sum of its two:
- (A) interior opposite angles
 - (B) base angles only
 - (C) adjacent interior angles
 - (D) three interior angles
11. Sewage is a complex mixture that contains suspended solids, dissolved impurities and many:
- (A) pure oxygen bubbles
 - (B) disease-causing microbes
 - (C) clean drinking water
 - (D) precious minerals for drinking
12. Find 25 percent of 200. The value works out to:
- (A) 100
 - (B) 50
 - (C) 25
 - (D) 75
13. In a plane mirror your right hand appears to be the left hand of the image. This effect is called:
- (A) lateral inversion
 - (B) dispersion
 - (C) refraction
 - (D) magnification
14. A triangle in which all three sides are equal is called:
- (A) equilateral
 - (B) scalene
 - (C) right-angled
 - (D) isosceles

15. The whole underground network of pipes that carries sewage from buildings to the treatment plant is the:

- (A) ventilation duct
- (B) water supply pipeline
- (C) sewerage system
- (D) electricity grid

16. The fraction $\frac{3}{4}$ written as a percentage is:

- (A) 43%
- (B) 34%
- (C) 7.5%
- (D) 75%

17. If a mirror's shiny surface caves inward, just like the inner bowl of a spoon, then it is a:

- (A) concave mirror
- (B) plane mirror
- (C) convex mirror
- (D) cylindrical mirror

18. A triangle with exactly two equal sides is called isosceles, and its two base angles are:

- (A) always 60°
- (B) unequal to each other
- (C) equal to each other
- (D) both right angles

19. The place where sewage is cleaned before the water is released back into a river is called a:

- (A) power station
- (B) petrol pump
- (C) reservoir
- (D) wastewater treatment plant

20. Written as a fraction in lowest terms, 40% equals:

- (A) $\frac{4}{10}$
- (B) $\frac{2}{5}$
- (C) $\frac{1}{4}$
- (D) $\frac{4}{5}$

21. The mirror used as a rear-view mirror in cars and on motorbikes is convex because it:

- (A) shows an upside-down image
- (B) forms a real image on the road
- (C) gives a smaller image and a wider field of view
- (D) magnifies the traffic behind

22. When every one of a triangle's three sides has a different length, the triangle is called:

- (A) right-angled
- (B) isosceles
- (C) equilateral
- (D) scalene

23. At the start of treatment, wastewater is passed through bar screens to remove:

- (A) dissolved salts
- (B) colour from the water
- (C) harmful microbes
- (D) large floating objects like rags and sticks

24. The decimal 0.6 expressed as a percentage is:

- (A) 600%
- (B) 0.6%
- (C) 60%
- (D) 6%

25. When white sunlight passes through a glass prism, it spreads out into a band of colours called the:

- (A) image
- (B) spectrum
- (C) shadow
- (D) reflection

26. The line segment joining a vertex of a triangle to the mid-point of the opposite side is called a:

- (A) hypotenuse
- (B) median
- (C) altitude
- (D) perpendicular bisector

27. After the bar screens, the water enters a grit and sand removal tank where the heavy particles:

- (A) float to the top
- (B) dissolve completely
- (C) turn into gas
- (D) settle down to the bottom

28. A shopkeeper buys a pen for ₹40 and sells it for ₹50. The profit percent is:

- (A) 10%
- (B) 25%
- (C) 20%
- (D) 50%

29. A rainbow in the sky is formed because tiny water drops act like prisms and cause:
- (A) reflection of sunlight with no colour change
 - (B) clouds glowing on their own
 - (C) dispersion of sunlight into seven colours
 - (D) the Sun changing colour
30. The perpendicular drawn from a vertex of a triangle to the line of its opposite side is the:
- (A) angle bisector
 - (B) median
 - (C) diagonal
 - (D) altitude
31. In the sedimentation tank, the solid waste that slowly settles at the bottom is called:
- (A) clarified water
 - (B) grit
 - (C) sludge
 - (D) scum
32. An article costing ₹500 is sold for ₹450. The loss percent is:
- (A) 10%
 - (B) 5%
 - (C) 90%
 - (D) 50%
33. The seven colours of the spectrum, in correct order, are remembered by the word VIBGYOR, which starts with:
- (A) green
 - (B) red
 - (C) violet
 - (D) white
34. How many medians does a triangle have?
- (A) two
 - (B) one
 - (C) six
 - (D) three
35. Floating oil, grease and other light material on top of the settling tank is removed by skimmers and is called:
- (A) sludge
 - (B) biogas
 - (C) scum
 - (D) clarified water

36. The simple interest on ■2,000 at 5% per year for 2 years is:
- (A) ■200
 - (B) ■400
 - (C) ■100
 - (D) ■2,200
37. A magnifying glass that makes letters look bigger when held close to a page is a:
- (A) convex lens
 - (B) flat glass sheet
 - (C) concave lens
 - (D) plane mirror
38. In a right-angled triangle, the side opposite the right angle is the longest side, called the:
- (A) median
 - (B) base
 - (C) hypotenuse
 - (D) altitude
39. The cleaner water left after the solids have settled and the scum is skimmed off is called:
- (A) sludge
 - (B) clarified water
 - (C) raw sewage
 - (D) drinking water
40. If 20 out of 50 students wear glasses, the percentage who wear glasses is:
- (A) 70%
 - (B) 50%
 - (C) 20%
 - (D) 40%
41. An image that can actually be caught on a screen is called a:
- (A) real image
 - (B) erect image
 - (C) laterally inverted image
 - (D) virtual image
42. By the Pythagoras property, in a right triangle with legs 3 cm and 4 cm, the hypotenuse is:
- (A) 6 cm
 - (B) 7 cm
 - (C) 5 cm
 - (D) 12 cm
43. Clarified water is cleaned further by pumping air into it so that helpful aerobic bacteria can:
- (A) kill all forms of life instantly
 - (B) freeze the water solid
 - (C) grow and consume the remaining wastes
 - (D) add colour to the water

44. A shirt marked ₦800 is sold at a 15% discount. The discount amount is:
- (A) ₦15
 - (B) ₦150
 - (C) ₦120
 - (D) ₦680
45. The straight line drawn at right angles to the mirror at the point where the ray hits is called the:
- (A) normal
 - (B) reflected ray
 - (C) incident ray
 - (D) axis of the mirror
46. The three sides of a triangle measure 6 cm, 8 cm and 10 cm. The hypotenuse-like longest side shows this is a:
- (A) right-angled triangle
 - (B) an impossible triangle
 - (C) equilateral triangle
 - (D) obtuse triangle
47. The settled sludge is taken to a tank where bacteria decompose it without air (anaerobically), producing a useful gas called:
- (A) petrol
 - (B) biogas
 - (C) chlorine
 - (D) oxygen
48. The ratio of boys to girls in a class is 3 : 2. If there are 30 students in all, the number of girls is:
- (A) 10
 - (B) 12
 - (C) 18
 - (D) 15
49. A periscope, used to see over a wall or out of a submarine, works using two:
- (A) glass prisms only
 - (B) convex lenses
 - (C) plane mirrors set at 45°
 - (D) concave mirrors
50. Which set of lengths can form a triangle? (Sum of any two sides must be greater than the third.)
- (A) 4 cm, 4 cm, 9 cm
 - (B) 5 cm, 6 cm, 9 cm
 - (C) 1 cm, 2 cm, 3 cm
 - (D) 2 cm, 3 cm, 6 cm

51. After it is dried, the decomposed sludge from a treatment plant can be used as:
- (A) drinking water
 - (B) manure
 - (C) building cement
 - (D) fuel for cars
52. 30% of a number is 60. The number is:
- (A) 200
 - (B) 90
 - (C) 180
 - (D) 18
53. When light falls on a rough, uneven surface and scatters in many directions, it is called:
- (A) refraction
 - (B) diffuse reflection
 - (C) regular reflection
 - (D) dispersion
54. Two angles of a triangle are equal and the third is 80° . Each of the two equal angles is:
- (A) 50°
 - (B) 40°
 - (C) 80°
 - (D) 100°
55. Throwing cooking oil and fats down the kitchen drain is harmful because they:
- (A) kill all the harmful microbes
 - (B) clean the pipes thoroughly
 - (C) turn into clean water
 - (D) harden and block the drains and pipes
56. The percentage 45% written as a decimal is:
- (A) 0.045
 - (B) 45.0
 - (C) 4.5
 - (D) 0.45
57. A concave mirror is used in a car headlight and a torch because it can:
- (A) split white light into colours
 - (B) send out a strong, parallel beam of light
 - (C) shrink the scene like a rear-view mirror
 - (D) spread light weakly in all directions
58. In a right-angled triangle one angle is 90° . The other two acute angles must add up to:
- (A) 90° each
 - (B) 180°
 - (C) 90°
 - (D) 45°

59. Used tea leaves, solid food and cotton should be thrown in the dustbin, not the drain, because they:
- (A) speed up the water flow
 - (B) clean the drains
 - (C) choke the drains
 - (D) become safe drinking water
60. The price of a cycle rises from ■2,000 to ■2,400. The percentage increase is:
- (A) 20%
 - (B) 40%
 - (C) 400%
 - (D) 16.6%
61. In a plane mirror, the distance of the image behind the mirror equals the distance of the object in front. So an object 30 cm in front forms an image that is:
- (A) 60 cm behind the mirror
 - (B) 30 cm in front of the mirror
 - (C) 30 cm behind the mirror
 - (D) 15 cm behind the mirror
62. An exterior angle of a triangle is 120° , and one interior opposite angle is 70° . The other interior opposite angle is:
- (A) 50°
 - (B) 60°
 - (C) 70°
 - (D) 120°
63. In places without a sewerage network, household wastewater is often treated on-site in an underground:
- (A) rainwater barrel
 - (B) petrol tank
 - (C) overhead water tank
 - (D) septic tank
64. Two ratios are equivalent if one can be obtained from the other by multiplying both terms by the same number. Which ratio is equivalent to 2 : 3?
- (A) 2 : 6
 - (B) 8 : 12
 - (C) 4 : 9
 - (D) 6 : 8
65. A concave lens is thinner in the middle than at the edges, and it always makes a distant object look:
- (A) smaller
 - (B) exactly the same size
 - (C) upside-down
 - (D) larger

66. The point where the three medians of a triangle meet is called the:
- (A) midpoint
 - (B) vertex
 - (C) hypotenuse
 - (D) centroid
67. Poor sanitation and open drains are dangerous mainly because stagnant dirty water lets:
- (A) electricity to be generated
 - (B) clean drinking water to form
 - (C) mosquitoes and disease germs breed
 - (D) fresh oxygen to fill the air
68. A bag has red and blue balls in the ratio 5 : 3. The fraction of the balls that are blue is:
- (A) $\frac{5}{8}$
 - (B) $\frac{3}{8}$
 - (C) $\frac{3}{15}$
 - (D) $\frac{3}{5}$
69. Two plane mirrors are placed facing each other; an object between them forms:
- (A) only a coloured band
 - (B) no image at all
 - (C) exactly one image
 - (D) many images
70. The perimeter of an equilateral triangle of side 7 cm is:
- (A) 28 cm
 - (B) 14 cm
 - (C) 21 cm
 - (D) 7 cm
71. A low-cost, on-site sanitation method in which earthworms turn human waste into compost is called a:
- (A) public fountain
 - (B) flush toilet linked to a sewer
 - (C) vermi-processing toilet
 - (D) rainwater harvesting pit
72. A man spends 70% of his salary and saves the rest. If he saves ₹6,000, his salary is:
- (A) ₹8,571
 - (B) ₹20,000
 - (C) ₹60,000
 - (D) ₹6,000

73. A ray of light strikes a plane mirror making an angle of 30° with the normal. The angle of reflection is:

- (A) 60°
- (B) 90°
- (C) 15°
- (D) 30°

74. A right triangle has a hypotenuse of 13 cm and one leg of 12 cm. The other leg is:

- (A) 5 cm
- (B) 1 cm
- (C) 17 cm
- (D) 25 cm

75. Diseases such as cholera, typhoid and dysentery spread most often through:

- (A) water contaminated by untreated sewage
- (B) eating freshly cooked food
- (C) clean, treated tap water
- (D) breathing fresh mountain air

76. The simple interest on ₦5,000 at 8% per year for 3 years is:

- (A) ₦6,200
- (B) ₦1,200
- (C) ₦400
- (D) ₦1,500

77. A ray hits a plane mirror with an angle of incidence of 40° (measured from the normal). The total angle between the incident ray and the reflected ray is:

- (A) 40°
- (B) 20°
- (C) 80°
- (D) 90°

78. A triangle that has one angle greater than 90° is called an:

- (A) equilateral triangle
- (B) obtuse-angled triangle
- (C) acute-angled triangle
- (D) right-angled triangle

79. Before treated water is released, it is often disinfected with chlorine or ozone, or passed through UV, in order to:

- (A) freeze the water
- (B) add minerals for taste
- (C) kill the remaining harmful microbes
- (D) make the water blue

80. To compare the ratios 2 : 3 and 3 : 4, you can write them with a common denominator. The larger ratio is:

- (A) they are equal
- (B) cannot be compared
- (C) 3 : 4
- (D) 2 : 3

81. When all seven colours of the spectrum are combined back together, they form:

- (A) black light
- (B) no light
- (C) green light
- (D) white light

82. Is it possible for any single triangle to contain two separate right angles?

- (A) No, because two right angles already total 180°
- (B) Yes, if the third angle is 0°
- (C) Yes, if the sides are equal
- (D) Yes, in a right-angled triangle

83. A treatment plant takes in 800 litres of sewage and the bar screens plus grit removal take out 200 litres of solids and grit. The fraction of the intake removed at this early stage is:

- (A) one-eighth ($1/8$)
- (B) one-third ($1/3$)
- (C) one-half ($1/2$)
- (D) one-quarter ($1/4$)

84. A quantity of 12 is what percent of 48?

- (A) 25%
- (B) 48%
- (C) 12%
- (D) 400%

85. An image formed by a plane mirror is described as virtual because it:

- (A) changes colour
- (B) is smaller than the object
- (C) cannot be caught on a screen
- (D) is upside-down

86. In an equilateral triangle, each interior angle measures:

- (A) 30°
- (B) 60°
- (C) 90°
- (D) 45°

87. During treatment, 600 litres of water comes in and 540 litres of clean water comes out. The percentage recovered as clean water is:

- (A) 54%
- (B) 90%
- (C) 60%
- (D) 9%

88. ■900 is to be divided between two friends in the ratio 4 : 5. The smaller share is:

- (A) ■225
- (B) ■450
- (C) ■500
- (D) ■400

89. The image in a plane mirror is laterally inverted, which means the letter 'b' would look like:

- (A) the letter 'q'
- (B) the letter 'p'
- (C) the letter 'b' unchanged
- (D) the letter 'd'

90. The two equal sides of an isosceles triangle are each 8 cm and the base is 5 cm. Its perimeter is:

- (A) 13 cm
- (B) 40 cm
- (C) 16 cm
- (D) 21 cm

91. A village septic tank starts the month holding 50 kg of sludge and ends with 80 kg. The percentage increase in sludge is:

- (A) 37.5%
- (B) 160%
- (C) 30%
- (D) 60%

92. A number is increased by 10% and the result is 110% of the original. If the original is 80, the new value is:

- (A) 70
- (B) 90
- (C) 8
- (D) 88

93. A beam of sunlight enters a dark room through a small hole and lights up floating dust along a perfectly straight line. This best shows that light:

- (A) travels in a straight line
- (B) is made of seven colours only when reflected
- (C) cannot pass through air
- (D) bends around corners

- 94.** Two sides of a triangle are 7 cm and 10 cm. The third side must be less than:
- (A) 70 cm
 - (B) 17 cm
 - (C) 27 cm
 - (D) 3 cm
- 95.** A town's plant treats sewage in the ratio 3 parts cleaned to 1 part sent back as sludge. Out of 1,200 litres treated, the volume that becomes sludge is:
- (A) 300 litres
 - (B) 900 litres
 - (C) 400 litres
 - (D) 1200 litres
- 96.** A juice is made by mixing concentrate and water in the ratio 1 : 4. To make 1,000 mL of juice, the amount of concentrate needed is:
- (A) 100 mL
 - (B) 200 mL
 - (C) 800 mL
 - (D) 250 mL
- 97.** A ray of light hits a plane mirror straight on, along the normal (angle of incidence 0°). It is reflected back:
- (A) straight back along the same path
 - (B) at ninety degrees to the path
 - (C) scattered in all directions
 - (D) split into seven colours
- 98.** One acute angle of a right-angled triangle is 35° . The other acute angle is:
- (A) 55°
 - (B) 65°
 - (C) 145°
 - (D) 45°
- 99.** A plant cleaned 250 litres of sewage on Monday and 300 litres on Tuesday. Out of these two days, the percentage cleaned on Tuesday was:
- (A) about 45.5%
 - (B) exactly 30%
 - (C) about 54.5%
 - (D) exactly 60%
- 100.** A student scored 18 marks out of 20 in a test. Her percentage score is:
- (A) 18%
 - (B) 80%
 - (C) 90%
 - (D) 9%